PHYS4055

Atomic physics and quantum optics

Week 9

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In these lectures, we will look into:

- Hyperfine structure, in more detail
- Isotope shift
- E1 matrix elements, Einstein A coefficients, lifetimes
- Allowed and forbidden transitions
- Stark shift and electric polarisability
- Zeeman shift — weak and strong fields

Hyperfine structure (hfs)

Hyperfine structure: $$H_{\rm hfs} = \sum_i^N h_{\rm hfs}(\mathbf{r}_i) = \sum_{kq} (-1)^q T^k_q U^k_{-q}$$

$$= \sum_q (-1)^q T^1_q U^1_{-q} + \sum_q (-1)^q T^2_q U^2_{-q} + \ldots$$

- $k = 1$ - nuclear magnetic dipole moment with magnetic field of electrons;
- $k = 2$ - nuclear electric quadrupole moment with gradient of electron electric field;
- $k = 3$ - nuclear magnetic octupole moment with magnetic field of electrons; .....

Due to coupling of *I* and *J*, the atomic state is defined in terms of the total angular moment *F* and its projection *M*, $|(IJ), FM\rangle$.

Energy shift due to hfs: $$\Delta E_{\rm hfs} = \langle (IJ, FM) | H_{\rm hfs} | (IJ), FM \rangle$$

$$= \ldots \quad \text{(this will be an exercise for you in the next assignment!)}$$

$$= (-1)^{I+J+F} \begin{Bmatrix} I & J & F \\ J & I & k \end{Bmatrix} \langle J \| T^k \| J \rangle \langle I \| U^k \| I \rangle$$

$$= \frac{1}{2} KA + \frac{1}{2} \frac{3K(K+1) - 4J(J+1)I(I+1)}{2I(2I-1)2J(2J-1)} B + \ldots$$

Hyperfine structure (hfs)

Here: $\begin{Bmatrix} a & b & c \\ d & e & f \end{Bmatrix}$ denotes a 6-j symbol,

$$K = F(F+1) - J(J+1) - I(I+1),$$

$A = \dfrac{\mu}{IJ}\langle JJ\,|\,T_0^1\,|\,JJ\rangle$ is magnetic dipole hyperfine constant,

$B = 2Q\langle JJ\,|\,T_0^2\,|\,JJ\rangle$ is the electric quadrupole hyperfine constant.

For $I \geq J$ there are $2J+1$ levels.

For $I < J$ there are $2I+1$ levels.

The energy *splittings* are found from the differences between the neighbouring energy shifts. For $J = 1/2$ there is just one splitting, with

$$\delta E_{\rm hfs} = \Delta E_{\rm hfs}(I+1/2) - \Delta E_{\rm hfs}(I-1/2) = (I+1/2)A$$

Hyperfine structure (hfs)

Remember:

- *vector* in electron space must point along *total electron angular momentum J ;*
- *vector* in nuclear space must point along *total nuclear angular momentum I .*

For magnetic dipole:

$$h_{\mathrm{hfs}} \propto \mathbf{I}\cdot\mathbf{J} = A\mathbf{I}\cdot\mathbf{J}$$

$$\Rightarrow \quad \Delta E_{\mathrm{hfs}} = \langle h_{\mathrm{hfs}}\rangle = A\langle \mathbf{I}\cdot\mathbf{J}\rangle = KA/2 \ .$$

$\langle I \| T^k \| I \rangle$

$\langle J \| T^k \| J \rangle$

$\begin{pmatrix} J_1 & k & J_2 \\ -m_1 & q & m_2 \end{pmatrix}$

$|J_1 - J_2| \le k \le J_1 + J_2$

Much simpler way to figure it out!

Different hyperfine structure for isotopes with different nuclear spin *I*.

Exercise: What are the angular momentum requirements for *I* and *J* in order to observe the effects of the nuclear moments through the hyperfine structure?

For mag. dipole to contribute ($k=1$): $I, J \geq \frac{1}{2}$

electric quadrupole $k=2$: $I, J \geq 1$

mag. octupole $k=3$: $I, J \geq 3/2$

Isotope shift

Consider an atom with a nucleus with protons Z and neutron N. There is often more than 1 possible isotope that may be experimentally studied. The *isotope shift* is the difference in the energies of the states between an isotope with neutrons N' and that with neutrons N.

There are two sources of this difference: the difference in the *masses* of the nuclei; and the difference in the *volume* of the nuclei.

Here, we will consider the most significant of these effects for heavy atoms — the "volume" or "field" shift. This change in the volume may be parametrized by the differential mean square nuclear charge radius, $\delta\langle r^2\rangle$,

$$\delta E = -F\delta\langle r^2\rangle \,.$$

Let's model the nuclear charge distribution as uniform. Then within the nuclear radius, $r \le R$,

$$V(r) = -\frac{Z}{R}\left[\frac{3}{2} - \frac{1}{2}\left(\frac{r}{R}\right)^2\right] .$$

Varying the potential with respect to the nuclear radius leads to

$$\delta V(r) = \frac{3Z}{2R^2}\left[1 - \left(\frac{r}{R}\right)^2\right]\delta R \,.$$

The nuclear radius and the mean square nuclear charge radius are related by $\langle r^2\rangle = (3/5)R^2$, so $\delta\langle r^2\rangle = (6/5)R\,\delta R$, and therefore

$$\delta V(r) = \frac{5Z}{4R^3}\left[1 - \left(\frac{r}{R}\right)^2\right]\delta\langle r^2\rangle = -F(r)\delta\langle r^2\rangle$$

with $F = \langle F(r)\rangle$.

$\delta E = \langle \delta V \rangle$
$= -\langle F(r) \rangle \delta\langle r^2\rangle$
$= -F\,\delta\langle r^2\rangle$

Absorption and emission of radiation

So far we have considered stationary atomic states. Now we will consider *transitions* between different atomic states due to the absorption or emission of electromagnetic radiation. We will touch on absorption and stimulated emission, and focus on spontaneous emission of radiation.

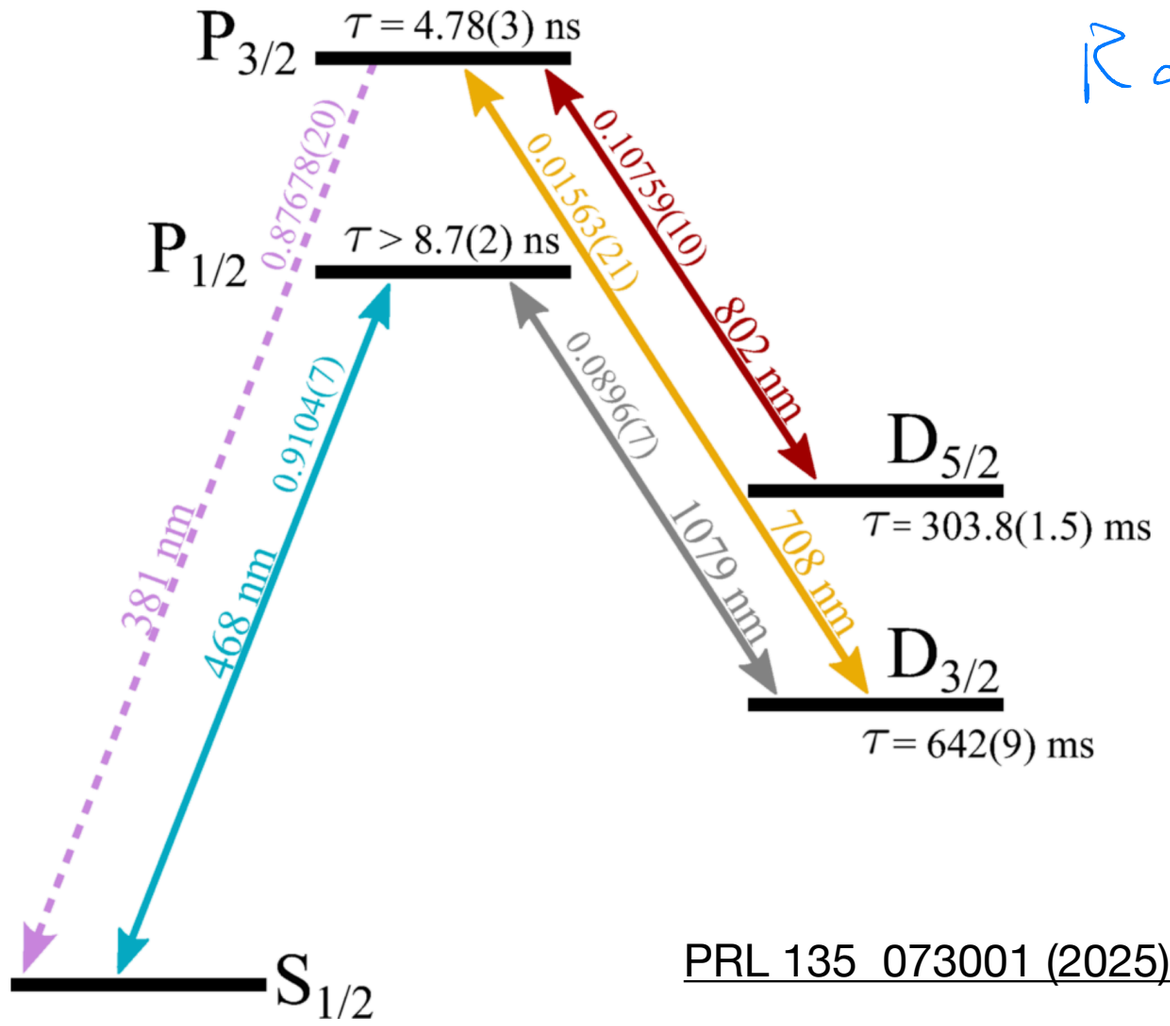
In these processes, we need to consider not only conservation of energy — with the photon adding or removing $\hbar\omega = |E_i - E_f|$ of energy — though also conservation of angular momentum. Therefore, the polarisation of the radiation will need to be taken into account.

<u>We would like to understand:</u> which are the transitions that are *allowed?*

That is, what are the quantum numbers that allow a transition to proceed or not, and what else is needed to determine the probability of the transition?

How do we determine the *lifetime of the state?*

E.g., lifetimes of the P states of the radium ion (right figure) are on the order of ns, while for the D states it is 10^7 times longer!



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Absorption and emission of radiation

Absorption

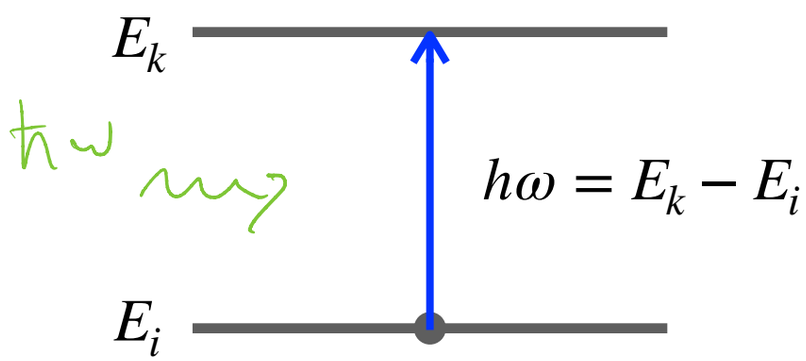
Consider a collection of atoms in equilibrium with the electromagnetic radiation field having *spectral density function* $\rho(\omega)$.

$\rho(\omega)$ — photon energy per unit volume in interval $d\omega$

Probability per second of *absorbing* a photon and leading to a transition of the atom from state *i* to state *k* is

$$w_{ik}^{\text{abs}} = B_{ik}\rho(\omega)$$

where B is the Einstein coefficient for absorption,

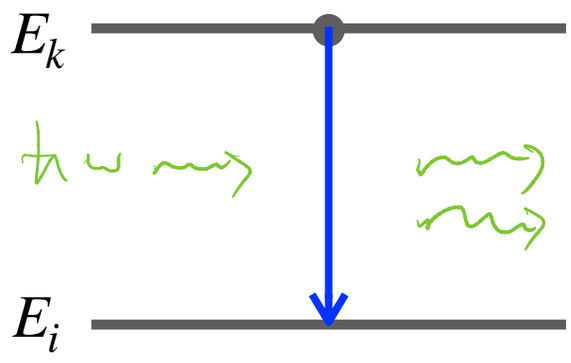


Absorption and emission of radiation

Induced or stimulated emission

The radiation field may also induce atoms in an excited atomic state *k* to emit a photon with energy $\hbar\omega = E_k - E_i$ and to drop to a lower state *i.*

Here, the photons have the same propagation directions.



The induced emission probability may be given by

$$w_{ki}^{\text{ind}} = B_{ki}\rho(\omega)$$

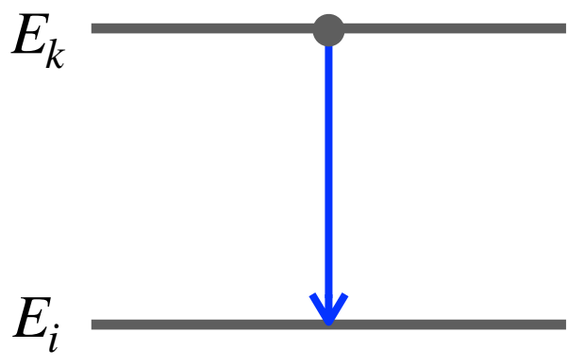
where B is the Einstein coefficient for induced emission.

Absorption and emission of radiation

Spontaneous emission

Even *without* the radiation field, an atom in an excited state *k* may drop to a lower state *i,* emitting a photon of energy $\hbar\omega = E_k - E_i$.

In this case, the photon may be emitted in any direction.



The probability for spontaneous emission is given br

$$w_{ki}^{\mathrm{sp}} = A_{ki}$$

where A is the Einstein coefficient for spontaneous emission.

This process depends only on the wave functions of the states *i* and *k* of the atom

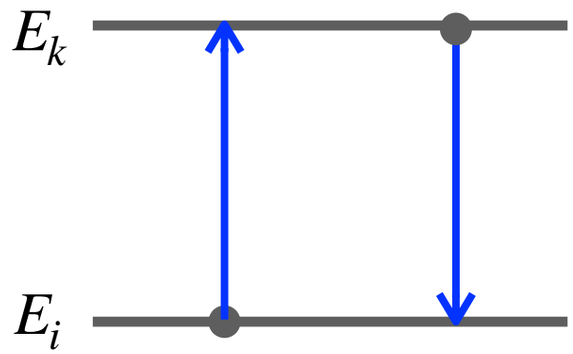
The lifetimes of atomic states are determined by this process

Absorption and emission of radiation

Relationship between Einstein coefficients:

$$g_k B_{ki} = g_i B_{ik}$$

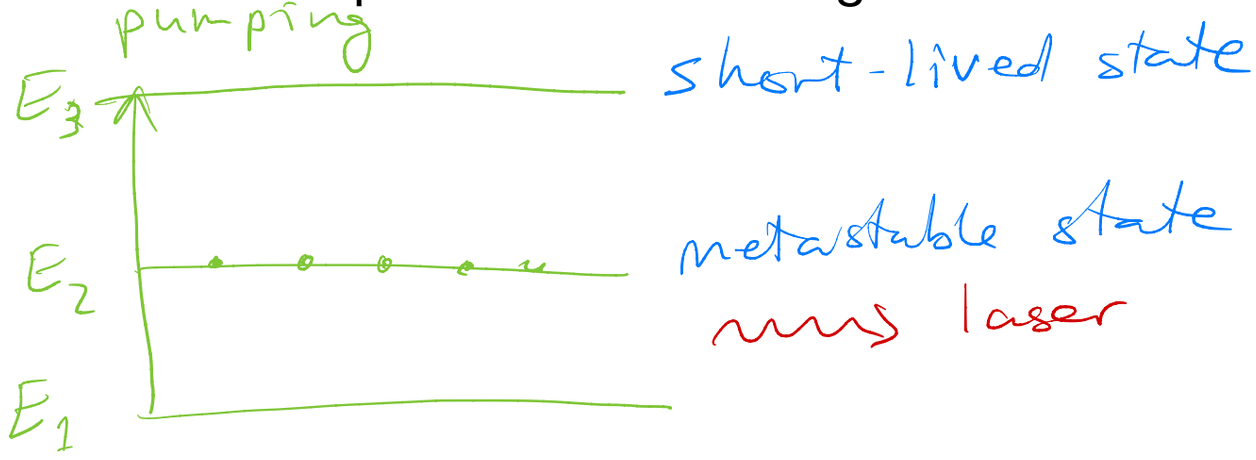
$$A_{ki} = \frac{\hbar\omega^3}{\pi^2 c^3} B_{ik}$$



g_i and g_k give the degeneracy of the states. E.g., $g_i = 2J_i + 1$, for the state *i* with total angular momentum J_i with magnetic substates with the same energy

For $g_i = g_k$, probability of absorption and induced emission is the same

Population inversion is required to make a laser. This corresponds to there being more atoms in the upper than the lower state



EM transitions

To determine atomic transition probabilities, time-dependence needs to be considered.

The transition amplitude from the initial to final states may be given by

$$T_{fi} = \langle \Psi_f | H_I | \Psi_i \rangle \qquad \text{for absorption of radiation}$$

$$T_{fi} = \langle \Psi_f | H_I^{\dagger} | \Psi_i \rangle \qquad \text{for emission of radiation}$$

where $H_I(t) = e^{iHt/\hbar} H_I e^{-iHt/\hbar}$ is the time-dependent interaction Hamiltonian .

The probability w_{fi} for a transition involves the square of the amplitude $|T_{fi}|^2$.

read through Johnson
if interested in derivation

E1 transitions

Electric dipole (E1)

The dominant transition is the electric dipole transition. It corresponds to the electric dipole operator $\mathbf{d} = e\mathbf{r}$, having matrix element

$$D_{fi} = \langle \varphi_f | D_z | \varphi_i \rangle = e\langle \varphi_f | z | \varphi_i \rangle$$

Allowed between states:

Reminder: going from cartesian to spherical tensor operators $\Rightarrow$ D_q $(q = 0, \pm 1)$

$$D_1 = -\frac{1}{\sqrt{2}}(D_x + iD_y)$$

$$D_0 = D_z$$

$$D_{-1} = \frac{1}{\sqrt{2}}(D_x - iD_y)$$

$\Rightarrow$ rank 1 tensor $\Rightarrow$ can change J by up to 1.

$$|J_i - J_f| \leq 1 \leq J_i + J_f$$

$$\Rightarrow \quad \Delta J = 0, \pm 1$$

Parity? must have opposite parity!

Opposite parity?

If non-rel. & single-electron states, then opp. parity is clear.

Parity is odd if $\ell = 1, 3, 5, \ldots$

even if $\ell = 0, 2, 4, \ldots$

So s & p have opp. parity.

$\Rightarrow$ parity $P = (-1)^\ell$

$\Rightarrow$ $P = -1$ odd

$P = +1$ even.

$$\psi_{n\ell m}(r, \theta, \varphi) = \underbrace{R_{n\ell}(r)}_{P_{\ell n}} Y_{\ell m}(\theta, \varphi)$$

Parity operator: $x \to -x$, $y \to -y$, $z \to -z$

$\Rightarrow$ in spherical coordinates: $r \to r$, $\theta \to \pi - \theta$, $\varphi \to \varphi + \pi$

$$Y_{\ell m}(\pi - \theta, \varphi + \pi) = (-1)^\ell Y_{\ell m}(\theta, \varphi)$$

Parity for more than 1 electron?

$$P = (-1)^{l_1} (-1)^{l_2} (-1)^{l_3} \cdots l^{l_N}$$

$$= (-1)^{\sum_{i=1}^{N} l_i}, \quad N \text{ electrons}$$

Note: $\sum_i l_i \neq$ total orbital angular momentum of state.

Question: what is the parity of

- $1s^2 2s^2 2p^6 3s$, $[Ne]3p$ ($l=1$) , $[Ne]3d$ $(-1)^2 = 1$ even

  $[Ne]$

  $(-1)^l = (-1)^0 = 1$ even

  $(-1)^1 = -1$ odd

  states of Na?

- $[Ne]3s^2$, $[Ne]3s3p$, $[Ne]3s3d$

  $(-1)^{l_1+l_2}$

  states of Mg?

  $(-1)^{0+0} = 1$ $(-1)^{0+1} = -1$ $(-1)^{0+2} = 1$

E1 transition probability.

Spontaneous emission, state $a \to b$.

S_{E1} – line strength.

$$A_{ab} = \frac{4\alpha}{3} \frac{\omega^3}{c^2} \frac{|\langle b \| r \| a \rangle|^2}{2j_a+1}$$

atomic units

$$= \frac{2.02613 \times 10^{18}}{\lambda^3} \frac{S_{E1}}{2j_a+1} \; s^{-1}.$$

λ in Å $(10^{-10}\,m)$.

M1 and E2 transitions

Magnetic dipole (M1)

Operator : $\underset{\sim}{M} = \frac{1}{2m}\left(\underset{\sim}{L} + 2\underset{\sim}{S}\right)$ [Note: no radial dependence]

Probability : $A_{ab} = \frac{1}{3}\frac{\omega^3}{c^5}\frac{|\langle b||\underset{\sim}{L}+2\underset{\sim}{S}||a\rangle|^2}{2j_a+1}$ (S_{M1})

$= \frac{2.69735\times10^{13}}{\lambda^3}\frac{S_{M1}}{2j_a+1}\ s^{-1}$.

$\int P_a^{(\alpha)} P_b\, dr = 0$.

Selection rules : $|j_a - 1| \le j_b \le j_a + 1$; same parity ; $n_a l_a \not\to n_b l_a$ forbidden in non-rel. treatment.

Electric quadrupole (E2)

Operator $Q_q^{(2)} = r^2 C_q^{(2)}(\underset{\sim}{r})$ $\quad C_q^{(2)} \propto Y_q^2(\underset{\sim}{r})$

Probability: $A_{ab} = \frac{1}{15}\frac{\omega^5}{c^5}\frac{|\langle b||Q^{(2)}||a\rangle|^2}{2j_a+1}$ (S_{E2})

$= \frac{1.1198\times10^{18}}{\lambda^5}\frac{S_{E2}}{2j_a+1}\ s^{-1}$.

Approx value for λ in optical transition)

$\sim 500\,nm = 5000\,\text{Å}$

cf. $\frac{1}{(5000)^5}$ to $\frac{1}{(5000)^3}$

Selection rules : $|j_a - 2| \le j_b \le j_a + 2$; same parity.

Lifetime of states

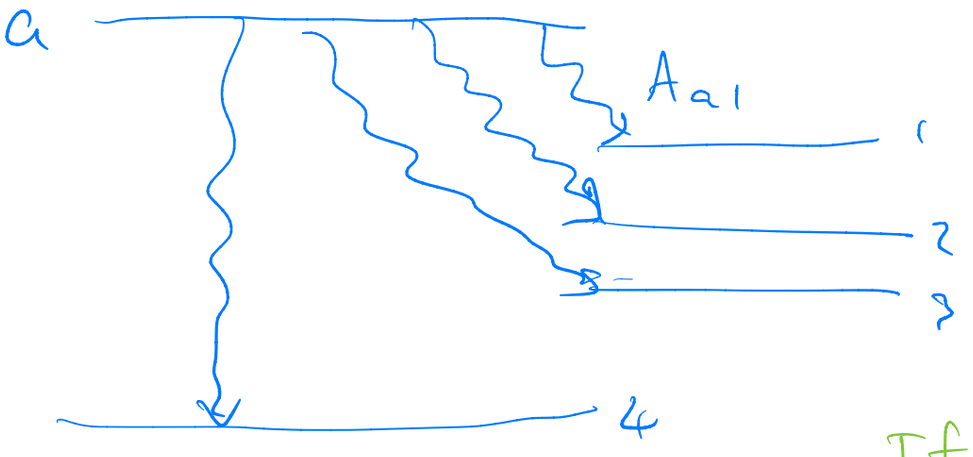
Total probability of decay from state a :

$$A_{a1} + A_{a2} + A_{a3} + \cdots$$

$$\sum_b A_{ab}$$

→ lifetime of state a is

$$\tau_a = \frac{1}{\sum_b A_{ab}}$$



$$\tau_a = \frac{1}{A_{a_1} + A_{a_2} + A_{a3} + A_{a4}}$$

If there are E1 transitions from state a, then don't need to consider smaller M1, E2.

Atom in external electric field: Stark shift

Perturbing interaction $H' = -\underset{\sim}{D}\cdot\underset{\sim}{\mathcal{E}} = \sum_i \underset{\sim}{r}_i \cdot \underset{\sim}{\mathcal{E}}$

$\underset{\sim}{D}$: electric dipole operator

$\sum_i$: over N electrons in atom

$\underset{\sim}{\mathcal{E}}$: external homogeneous electric field.

Choose z-axis to be along electric field direction:

$$\Rightarrow \quad H' = -D_z \mathcal{E}$$

First-order correction $\Delta E^{(1)} = \langle \gamma JM | H' | \gamma JM \rangle$

$$= -\langle \gamma JM | D_z | \gamma JM \rangle \mathcal{E}.$$

$= 0$ due to selection rules

$\Rightarrow$ Energy shift in the 2nd order

$$\Rightarrow \Delta E_{\gamma JM} = \sum_{\gamma' J'} \frac{|\langle \gamma JM | D_z | \gamma' J' M \rangle|^2}{E_{\gamma J} - E_{\gamma' J'}} \varepsilon^2$$

By evaluating the matrix elements it is found that

$$\langle \gamma JM | D_z | \gamma' J' M \rangle \propto \begin{cases} (J^2 - M^2)^{\frac{1}{2}}, & J' = J-1 \\ M, & J' = J \\ (J^2 - M^2)^{\frac{1}{2}}, & J' = J+1 \end{cases}$$

$\Rightarrow$ the levels shift and split* with two-fold† degeneracy (states with projections M & $-M$ have the same energy):

$$\Rightarrow \Delta E_{\gamma JM} = (a_{\gamma J} + b_{\gamma J} M^2) \varepsilon^2$$

($a_{\gamma J}$, $b_{\gamma J}$: constants)

* Except for levels with $J = 0$ & $J = \frac{1}{2}$.

† Except for level $M = 0$.

Atom in external magnetic field: Zeeman shift

The application of a magnetic field completely removes the degeneracy of levels w.r.t. M, unlike an electric field.

The interaction of an atom with a magnetic field is given by

$$H_{ZE} = -\underset{\sim}{\mu} \cdot \underset{\sim}{B}$$

($\underset{\sim}{\mu}$: magnetic moment of atom; $\underset{\sim}{B}$: external magnetic field.)

Consider LS coupling, and $\Delta E_{ZE} \ll \Delta E_{SO}$.

$\Rightarrow$ Expectation value in the basis $|\gamma L S J M\rangle$.

The operator of the magnetic moment of an electron is

$$\underset{\sim}{\mu} = -\mu_B \left(g_l \underset{\sim}{l} + g_s \underset{\sim}{s} \right), \quad \text{with } g_l = 1,\ g_s \approx 2.$$

($\underset{\sim}{l}$: orbital; $\underset{\sim}{s}$: spin)

The magnetic moment of the atom in state γJ is:

$$\underset{\sim}{\mu} = -\mu_B g_J \underset{\sim}{J}.$$

Here we have omitted the contribution from the nucleus, as it is orders of magnitude smaller than the electronic part ($\mu_N \ll \mu_B$)

$$\Rightarrow g_J \underset{\sim}{J} = g_L \sum_i \underset{\sim}{l}_i + g_S \sum_i \underset{\sim}{s}_i$$

$$= \underset{\sim}{L} + 2\underset{\sim}{S} = \underset{\sim}{J} + \underset{\sim}{S}.$$

$$\Rightarrow (g_J - 1)\underset{\sim}{J} = \underset{\sim}{S}.$$

$$\Rightarrow (g_J - 1)\langle \underset{\sim}{J}\cdot\underset{\sim}{J}\rangle = \langle \underset{\sim}{S}\cdot\underset{\sim}{J}\rangle$$

$$\Rightarrow (g_J - 1)\langle J^2\rangle = \langle \underset{\sim}{S}\cdot\underset{\sim}{S}\rangle + \langle \underset{\sim}{S}\cdot\underset{\sim}{L}\rangle$$

$$\Rightarrow (g_J - 1)\langle J^2\rangle = S(S+1) + \frac{1}{2}\times$$
$$\left[J(J+1) - S(S+1) - L(L+1)\right]$$

$$= \frac{1}{2}\left[J(J+1) + S(S+1) - L(L+1)\right]$$

$$\Rightarrow g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

$$\Rightarrow g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}$$

$$\Rightarrow H_{ZE} = \mu_B g_J \frac{\vec{J}}{\hbar} \cdot \frac{\vec{B}}{\hbar} = \mu_B g_J M_J B$$

This is the situation where nuclear spin $I=0$, or the hyperfine interaction is neglected.

Zeeman effect and hyperfine structure:

We consider two cases:

Weak field: $\Delta E_{ZE} \ll \Delta E_{HFS}$

Here, state is defined by

$$|\gamma I J F M_F\rangle,$$

F, M_F are good quantum numbers, not M_S or M_L.

And any vector in the system is directed along $\underset{\sim}{F}$ (not $\underset{\sim}{J}$, as before).

$$\Rightarrow \quad \underset{\sim}{\mu} = -\mu_B g_F \underset{\sim}{F} = -\mu_B g_J \underset{\sim}{J}.$$

$$\Rightarrow \quad g_F \langle \underset{\sim}{F} \cdot \underset{\sim}{F} \rangle = g_J \langle \underset{\sim}{J} \cdot \underset{\sim}{F} \rangle.$$

$$\Rightarrow \quad g_F = g_J \frac{\langle \underset{\sim}{J} \cdot \underset{\sim}{F} \rangle}{F(F+1)}$$

$$\Rightarrow \quad H_{ZE} = -\underset{\sim}{\mu} \cdot \underset{\sim}{B} = \mu_B g_F \underset{\sim}{F} \cdot \underset{\sim}{B}$$

$$= \mu_B g_F F_z B$$

(B defines z-axis).

$$g_F = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} g_J$$

$\Rightarrow$ Zeeman shift: $\Delta E_{ZE} = \mu_B g_F B M_F$

Splits into $2F+1$ levels.

Strong field : $\Delta E_{ZE} >> \Delta E_{HFS}$

F no longer a good quantum number. Define state by $|\gamma I M_I J M_J\rangle$, and include hyperfine interaction perturbatively:

magnetic hyperfine constant

$$\Delta E_{ZE} = \mu_B g_J M_J B + A \langle \underset{\sim}{I} \cdot \underset{\sim}{J} \rangle$$

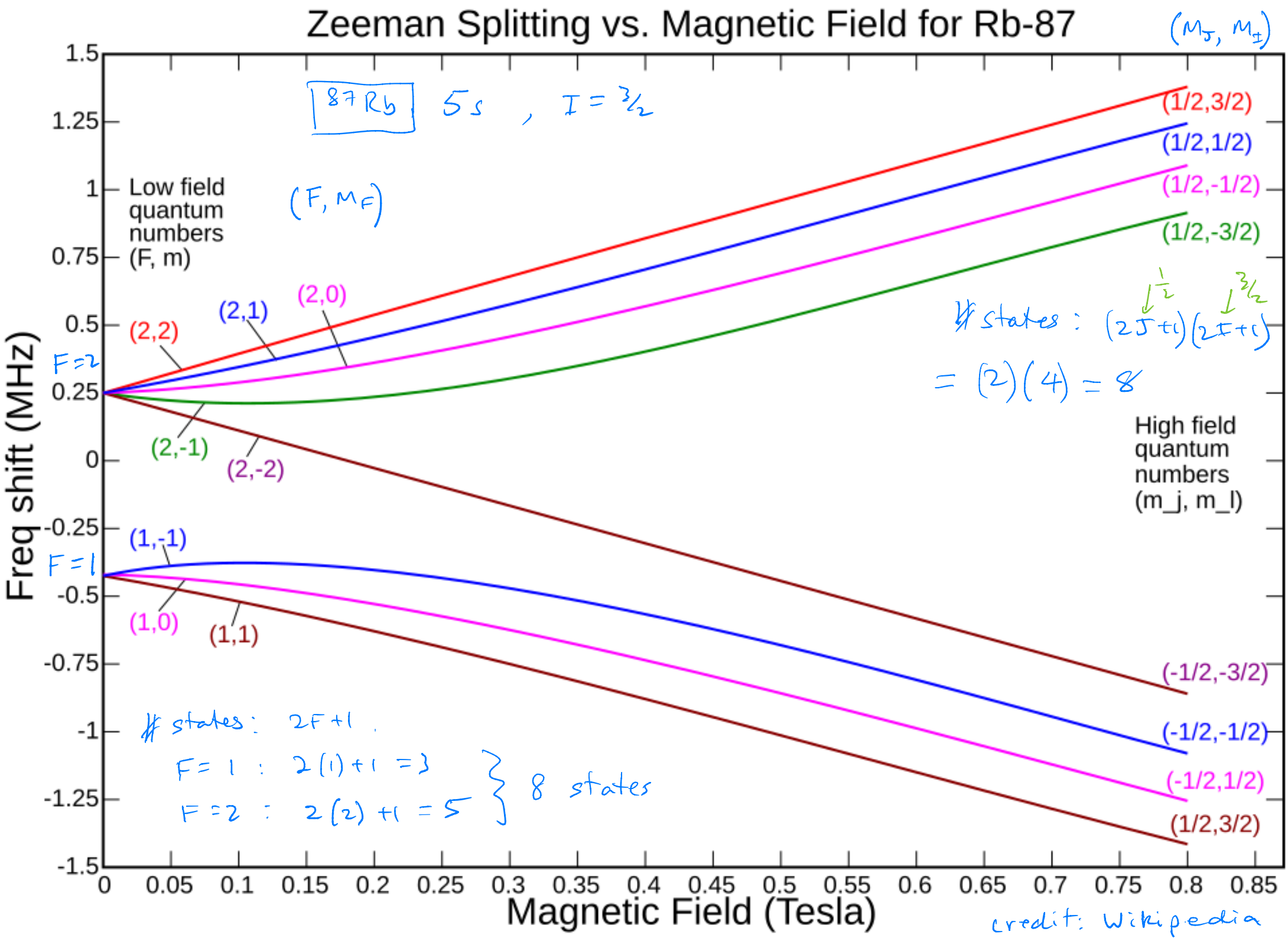
averaged over state $|\gamma I M_I J M_J\rangle$

$$= \mu_B g_J M_J B + A M_I M_J$$

Note: x & y components of

$$\underset{\sim}{I} \cdot \underset{\sim}{J} = I_x J_x + I_y J_y + I_z J_z$$

average to zero around z-axis, $\langle \quad \rangle$.


Zeeman Splitting vs. Magnetic Field for Rb-87
(M_J, M_I)
87Rb 5s , I = 3/2
Low field quantum numbers (F, m)
(F, M_F)
(2,2)
(2,1)
(2,0)
(2,-1)
(2,-2)
F=2
(1,-1)
(1,0)
(1,1)
F=1
(1/2,3/2)
(1/2,1/2)
(1/2,-1/2)
(1/2,-3/2)
(-1/2,-3/2)
(-1/2,-1/2)
(-1/2,1/2)
(1/2,3/2)
states: (2J+1)(2I+1)
= (2)(4) = 8
High field quantum numbers (m_j, m_I)
states: 2F+1
F=1 : 2(1)+1 = 3
F=2 : 2(2)+1 = 5
} 8 states
Freq shift (MHz)
Magnetic Field (Tesla)
credit: Wikipedia

After these lectures, you should understand

- the mechanisms leading to the hyperfine structure and the isotope shift

- E1 transitions, Einstein A coefficients, lifetimes

- allowed and forbidden transitions

- Stark and Zeeman shifts

and be able to do theory manipulations related to these